MATH

Modular Tensor Category: with Applications in CFT, TQFT

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Abstract. This is a note on the topic of Modular Tensor Categories. It is mainly based on the discussion in [1]

Contents

1 Preliminaries

This section we introduce some concepts in Algebra and Representation that may help us understand category as a more general mathematical structure.

1.1 Intertwiner and Representation

Let G be a group and let (V, ρ), (W, σ) be two representations of G. A **intertwiner** (or **morphism of representations**) is a linear map

$$\varphi : V \to W \tag{1}$$

such that for all $g \in G$,

$$\varphi \circ \rho(g) = \sigma(g) \circ \varphi, \tag{2}$$

i.e., the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{\rho(g)} & V \\ \varphi\downarrow & & \downarrow\varphi \\ W & \xrightarrow[\sigma(g)]{} & W \end{array} \tag{3}$$

The set of all intertwiners from (V, ρ) to (W, σ) is denoted $\mathrm{Hom}_G(V, W)$. If φ is an isomorphism, the two representations are said to be **equivalent** (or **isomorphic**), written $(V, \rho) \simeq (W, \sigma)$.

Theorem 1.1 **Schur's Lemma**

Let (V, ρ) and (W, σ) be irreducible representations of G, and let $\varphi \in \mathrm{Hom}_G(V, W)$.
1. If $(V, \rho) \not\simeq (W, \sigma)$, then $\varphi = 0$.
2. If $V = W$ and G acts over $\mathbb{C}$, then $\varphi = \lambda \mathrm{id}_V$ for some $\lambda \in \mathbb{C}$.

This is a version of Shur's Lemma using the language of intertwiners.

1.1.1 Intertwiners of Lie Algebras

Now let $\mathfrak{g}$ be a Lie algebra and let (V, ρ), (W, σ) be representations of $\mathfrak{g}$, i.e., Lie algebra homomorphisms

$$\rho : \mathfrak{g} \to \mathfrak{gl}(V), \quad \sigma : \mathfrak{g} \to \mathfrak{gl}(W). \tag{4}$$

A **intertwiner of Lie algebra representations** is a linear map $\varphi : V \to W$ satisfying

$$\varphi \circ \rho(X) = \sigma(X) \circ \varphi, \quad \forall X \in \mathfrak{g}. \tag{5}$$

Equivalently, φ intertwines the $\mathfrak{g}$-actions:

$$\varphi(X \cdot v) = X \cdot \varphi(v), \quad \forall X \in \mathfrak{g},, v \in V, \tag{6}$$

where $X \cdot v := \rho(X)(v)$.

The category of representations of $\mathfrak{g}$ with intertwiners as morphisms is denoted $\mathrm{Rep}(\mathfrak{g})$.

Example

Let $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$ with standard generators e, f, h satisfying $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$. The inite-dimensional irreducible representations V_n $(n \in \mathbb{N})$ have basis $\{v_0, v_1, ..., v_n\}$ with

$$h \cdot v_k = (n - 2k)v_k, \quad e \cdot v_k = (n - k + 1)v_{k-1}, \quad f \cdot v_k = (k + 1)v_{k+1}. \tag{7}$$

By Schur's Lemma, $\mathrm{Hom}_{\mathfrak{sl}_2}(V_m, V_n) = 0$ for $m \neq n$, and $\mathrm{End}_{\mathfrak{sl}_2}(V_n) \simeq \mathbb{C}$.

1.1.2 CVO as Intertwiners

The Chiral Vertex Operators constructed in Moore and Seiberg's work can be understood as king of a generalization of the concept of intertwiners of representation of Virasoro Algebra (in fact any Chiral Algebra(eg. Kac Moody Algebra) can be used here).

See [2] for a constructive definition of CVOs. In the constructive definition, we are made believe that the CVOs are defined to compute the BPZ conformal blocks of RCFT (at least for ones on a plane and on a torus).

There is also an axiomatic definition of CVOs as a intertwining object between:

$$\begin{pmatrix} i \\ jk \end{pmatrix}(z) : \mathcal{A} \otimes \mathcal{A} \to \mathcal{A} \tag{8}$$

where $\mathcal{A}$ is a representation of the chiral algebra of CFT, eg. Virasoro Algebra. However, instead of following Equation 6 as a normal lie algebra, it satisfy a more complicated intertwining relation:

We frist define a "co-product" of the chiral algebra representation:

$$\begin{aligned}
&\Delta_z : \mathcal{A} \to \mathcal{A} \otimes \mathcal{A}, \\
&\Delta_z(L_n) = 1 \otimes L_n + \sum_{k=0}^{\infty} \binom{n+1}{k} z^{n+1-k} L_{k-1} \otimes 1.
\end{aligned} \tag{9}$$

This kinds of bizzare co-product in fact follows directly from Cauchy theorem of complec integral on with 3 point instertion of the virasoro generators.

Then we define CVO of intertwining objects satisfying:

1. **Intetwining** compatible to the co-product:

$$\begin{aligned}
(L_n)\binom{i}{jk}(z)(\beta \otimes \gamma) &= \binom{i}{jk}(z)(\Delta_z(L_n)(\beta \otimes \gamma)) \\
&= \binom{i}{jk}(z)\left[(\beta \otimes L_n \gamma) + \sum_{k=0}^{\infty} \binom{n+1}{k} z^{n+1-k} (L_{k-1}\beta \otimes \gamma)\right]
\end{aligned} \tag{10}$$

1. **Position dependence**:

$$\partial_z \binom{i}{jk}(z)(\beta \otimes \gamma) = \binom{i}{jk}(z)(L_{-1}\beta \otimes \gamma) \tag{11}$$

This abstract definition of CVO is equivalent to the constructive definition;

$$\binom{i}{jk}(z)(\beta \otimes *) = \Phi_{ik}^{j,\beta}(z) \tag{12}$$

where we view $\Phi_{ik}^{j,\beta}(z)$ as kinds of a chiral primary field with insertion at point z with a field in j representation and labeled by β.

To understand the mathematical structure of CFT, we commonly say call the space of CVOs with fixed i, j, k representation as V_{jk}^i and this is in fact by this definition:

- Mathematically, V_{jk}^i is the space of intertwiners of representations of chiral algebra with the above intertwining relation.
- Semiphysically, V_{jk}^i is the space of chiral primary fields with fixed representation i, j, k and the above position dependence and conformal transformation property.
- Physically, V_{jk}^i is the space of all possible conformal blocks with fixed representation i, j, k and position $0, z, \infty$.

Naively, we may think that the space V_{jk}^i is a 1 dimensional vector space due to the fact that the 3 point blocks is fully fixed by conformal symmetry. This is true for Virasoro Minimal Models, yet for general Kac Moody Theory, this space may be of higher dimension.

We often notation $N_{jk}^i \coloneqq \dim V_{jk}^i$ as the fusion rule of the CFT, which counts the number of independent conformal blocks with fixed representation i, j, k and position $0, z, \infty$.

1.2 Vertex Operator Algebras

2 Modular Tensor Categories

2.1 Category Basics

A Category $\mathcal{C}$ is a mathematical structure with 2 typse of data:

- Objects: $\mathrm{Obj}(\mathcal{C})$, for example, $U, V \in \mathrm{Obj}(\mathcal{C})$.
- Morphisms: for any two objects $U, V \in \mathrm{Obj}(\mathcal{C})$, there is a set of morphisms $\mathrm{Hom}(U, V)$, for example, $f \in \mathrm{Hom}\ (U, V)$.

We have much to say about some basic properties we assume:

1. **Composition** of morphisms: for any $f \in \mathrm{Hom}(U, V)$ and $g \in \mathrm{Hom}(V, W)$, there is a morphism $g \circ f \in \mathrm{Hom}(U, W)$.
2. **Identity Exist**: for any $U \in \mathrm{Obj}(\mathcal{C})$, there is a special morphism $\mathrm{id}_U \in \mathrm{Hom}(U, U)$, called the **identity morphism** of U, it satisfies:

$$\mathrm{id}_V \circ f = f = f \circ \mathrm{id}_U, \quad \forall f \in \mathrm{Hom}(U, V). \tag{13}$$

1. **Completeness** The set $\mathrm{Obj}(\mathcal{C})$ is complete under direct sum, i.e., for any $U, V \in \mathrm{Obj}(\mathcal{C})$, there is an object $U \oplus V \in \mathrm{Obj}(\mathcal{C})$.

Under this general definition for a quite general Category, we now can introduce more and more structures on it to finally construct a Modular Tensor Category, which is extensively used in the study of Conformal Field Theory and Topological Quantum Field Theory.

2.2 Modular Tensor Category

2.2.1 Abelian Category

An **Abelian Category** means:

1. There is a 0 Object $0 \in \mathrm{Obj}(\mathcal{C})$;
2. Morphisms possess many properties as follows:

- Every morphism set $\mathrm{Hom}(A, B)$ is an **abelian group**, that means that every morphism have structure of:
  - addition: for any $f, g \in \mathrm{Hom}(A, B)$, there is a morphism $f + g \in \mathrm{Hom}(A, B)$;
  - zero morphism: there is a morphism $0 \in \mathrm{Hom}(A, B)$ such that $f + 0 = f$ for all $f \in \mathrm{Hom}(A, B)$;
  - additive inverse: for any $f \in \mathrm{Hom}(A, B)$, there is a morphism $-f \in \mathrm{Hom}(A, B)$ such that $f + (-f) = 0$.
  - Abelian property: for any $f, g, h \in \mathrm{Hom}(A, B)$, we have $f + (g + h) = (f + g) + h$ and $f + g = g + f$.
- composition of morphisms is bilinear;

$$\begin{gathered} f \circ (g + h) = f \circ g + f \circ h, \quad (f + g) \circ h = f \circ h + g \circ h, \\ \forall f, g \in \mathrm{Hom}(B, C), \forall h \in \mathrm{Hom}(A, B). \end{gathered} \tag{14}$$

- Every morphism has a **kernel** and a **cokernel**

Note **Kernel and Cokernel**

Given a morphism $f : A \to B$ in $\mathcal{C}$:

- The **kernel** of f is an object K together with a morphism $\iota : K \to A$:

$$\ker(f) = (K, \iota) \tag{15}$$

such that $f \circ \iota = 0$ (0 is assume by the abelian structure), and which is universal with this property: for any Z and any $\varphi : Z \to A$ with $f \circ \varphi = 0$, there exists a unique morphism $u : Z \to K$ with $\iota \circ u = \varphi$.

- The **cokernel** of f is an object Q together with a morphism $\pi : B \to Q$:

$$\operatorname{coker}(f) = (Q, \pi) \tag{16}$$

such that $\pi \circ f = 0$, and which is universal: for any Z and any $\psi : B \to Z$ with $\psi \circ f = 0$, there exists a unique $v : Q \to Z$ with $v \circ \pi = \psi$.

- Every monomorphism is the (morphism part of) kernel of its cokernel, and every epimorphism is the cokernel of its kernel:

$$h = \ker(\operatorname{coker}(h)) \quad g = \operatorname{coker}(\ker(g)) \tag{17}$$

This means monomorphisms and epimorphisms are "as well-behaved as injections and surjections of vector spaces."

Note **Monomorphisms and Epimorphisms**

A morphism $h : A \to B$ is a **monomorphism** if $h \circ \alpha = h \circ \beta$ implies $\alpha = \beta$ (left-cancellable), and an **epimorphism** if $\alpha \circ h = \beta \circ h$ implies $\alpha = \beta$ (right-cancellable).

- Every morphism f admits a factorization $f = h \circ g$, where h is a monomorphism and g is an epimorphism.

The above are tedious and boring mathematical construction, for physicist is better to understand as:

- Morphisms are sort of "Linear maps" between objects, and the composition is just the composition of linear maps.

2.2.2 Tensor Category

A tensor category means that there is a tensor product $\otimes$ on the category defined both for objects and morphisms, such that:

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Bibliography

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[2] G. W. Moore and N. Seiberg, "Lectures on RCFT," in *1989 Banff NATO ASI: Physics, Geometry and Topology*, 1989, pp. 1–129.